

TIAN Zhuang

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AGENT SKILLS

- **Computer Agentic Skills:**
 - ✧ Full-stack engineering developer: Front-End (Vue, React), Back-End (Django, FastApi, Solidity...)
 - ✧ Database: MySQL, MongoDB, PostgreSQL, Redis, Spark, Docker(K8s) ...
 - ✧ Cloud server: Google Cloud (GCP), Hostinger...
 - ✧ Familiar with Python, Transformer and Mainstream Artificial Intelligence algorithms and Framework: LoRA, Langchain, LangGraph, XGboost, LLaMA, Q learning...
 - ✧ Data Visualization: Scikit-learn, NumPy, SciPy, Pandas and Matplotlib, Dashboard (Tableau)
 - ✧ Development tools: VSCode, OpenClaw, Qt framework, Anaconda, DevEco, Github and Agent Skills...
- **Languages:** IELTS (6.5), Mandarin (Putonghua Proficiency Test: Level 2-A)

EDUCATION

Lingnan University

2025 - 2026

Master of Science in Artificial Intelligence and Business Analysis

• Curriculum



- ✧ Practical Application of Deep Learning
- ✧ Business Data Management
- ✧ Data Visualization
- ✧ Principle of Data Analytics and Programming
- ✧ Programming with Generative AI
- ✧ Big Data Analytics
- ✧ Computer Vision
- ✧ Machine Learning for Business
- ✧ Blockchain

- **GPA:** 3.6/4.0 (Semester1)

Federation University Australia - Shaoguan University (3+1)

09/2021-06/2025

Bachelor of Engineering in Computer Science and Technology

Introduction: The undergraduate program is a "3+1" programme run by SGU in collaboration with Federation University Australia. Without going to Australia in the fourth year, I studied at SGU and received major courses provided by teachers from the Federation University Australia.

- **GPA:** 3.49/5.0 (84.94/100)
- **Certificate:** Certificate of National Information Security Test Program (Level 1)
- **Chinese Teacher Qualification Certificate:** Information Technology (Junior High School)
- **Honours & Awards:**
 - ✧ **China National Scholarship** (11/2024)
 - ✧ First-class Academic Scholarship of the School of Information Engineering (10/2024)
 - ✧ Merit Student of the School of Information Engineering (10/2024)
 - ✧ Outstanding Individual in Research Innovation (10/2024)
 - ✧ Academic Excellent Award (10/2024)

PROJECTS

Lingnan University Policy System Using Agentic Retrieval-Augmented Generation (RAG)Framework

12/2025

Background: The policy documents of Lingnan University are usually lengthy and complex, covering languages such as Cantonese, English and simplified Chinese. Traditional RAG has a lot of illusions, and the existing framework is difficult to fine-tune many details. So, I developed a custom Agentic RAG framework as required by my mentor.

GitHub link: <https://github.com/terzotian/RAGagent>

Web URL: <https://agentrag.app>

- Traditional question-answering (QA) systems, which rely on predefined QA pairs or keyword-based retrieval, have limited capacity to interpret complex or novel queries. While Large Language Models (LLMs) can generate coherent, human-like responses, they are prone to hallucinations—producing content not grounded in the source material. In policy and compliance contexts, where factual accuracy and transparency are essential, such errors are unacceptable. Retrieval-Augmented Generation (RAG) offers an effective solution by integrating retrieval and generation components. The retrieval module extracts relevant policy segments from a document repository, while the generation module synthesizes coherent responses grounded in those sources. This hybrid approach enhances factual reliability, mitigates hallucinations, and ensures that outputs remain verifiable and consistent with authoritative policy content.

Project Description:

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- This project aims to design, develop, and deploy a Policy Question Answering (QA) System leveraging the Agentic Retrieval-Augmented Generation (RAG) framework, fully implemented within the Google Cloud Platform (GCP) environment. The system will be built and evaluated using Lingnan University's policies, enabling users to query university policy documents and receive concise, accurate, and evidence-based responses. The primary objective of this project is developing an AI-driven, cloud-native application that integrates modern machine learning and cloud computing technologies.
- All components—including data storage, retrieval, inference, and deployment—will be designed, developed, tested, and managed entirely within the GCP ecosystem.
- Core Task:
 - ✧ Policy document processing on GCP
 - ✧ Semantic search and query retrieval
 - ✧ RAG model development and integration
 - ✧ RAG model evaluation and performance optimization
 - ✧ Deployment and service delivery on GCP
- Techniques that Involved:
 - ✧ Programming Skills
 - ✧ Knowledge of NLP and Machine Learning
 - ✧ Familiarity with Cloud Services
- Achievement:
 - ✧ Development of an RAG system
 - ✧ Practical experience with NLP pipelines, vector search, and document intelligence
 - ✧ Techniques for evaluating and mitigating hallucinations in generative AI systems
 - ✧ Hands-on training with GCP AI tools
- Remarks:
 - ✧ This project is highly practical in nature, with a strong focus on production launch.
 - ✧ Students will develop skills that are directly relevant to the Hong Kong job market.
 - ✧ External support from experienced AI technicians is also available.

SRR-Case-Processing-System

11/2025

Introduction: The Architecture Services Department (ArchSD) in Hong Kong aims to develop an AI-powered support system to enhance the efficiency of slope and tree maintenance workflows.

Overview: An AI-powered document processing system that automatically extracts and classifies case data from Slope Risk Reports (SRR) in multiple file formats. Transform unstructured documents (TXT, TMO PDFs, RCC PDFs) into standardized case records with AI-powered classification and data extraction.

GitHub link: <https://github.com/SRR-Project-Team/SRR-Case-Processing-System>

Web URL: Private

Key Features:

- ✧ Multi-format Processing: TXT files, TMO PDFs, RCC PDFs
- ✧ AI Classification: 92% accuracy case type, 98% subject matter
- ✧ Historical Case Matching: Search 5,298 historical cases for patterns
- ✧ OCR Processing: Advanced text recognition for scanned documents
- ✧ Batch Processing: Handle multiple files simultaneously
- ✧ Integrated Chat Interface: Popup-based file management
- ✧ Tree Information Tracking: 32,405 trees with location data
- ✧ Location-Slope Learning: 403 auto-learned mappings

This project participated in the 2026 Lingnan Cup, in conjunction with business report analysis, etc

Multi-Agents LLM Financial Trading Decision Making Framework (Enhance TradingAgents)

02/2026

Project Description: Built on large language models (LLMs) and multi-agent architecture, this end-to-end automated stock investment research and decision-making system realizes full-process automation from target analysis, multi-perspective debate, trading decision-making to risk control, ultimately generating structured trading signals (BUY/SELL/HOLD).

- The overall system architecture and split multi-agent roles (Market Analyst, Fundamental Researcher, Trader, Risk Controller, Debate Referee) to enhance decision robustness and interpretability through cross-role collaboration.
- Implemented the investment research decision workflow orchestration based on LangGraph, constructing a complete execution graph: Data Acquisition → Multi-dimensional Analysis → Bull-Bear Debate → Trading Proposal → Risk Validation → Signal Output.
- Established a unified LLM adaptation layer, supporting seamless switching between multi-vendor models (OpenAI / Google / Anthropic / Ollama) to improve system compatibility and scalability.
- Encapsulated a financial data toolchain, integrating data sources such as yfinance and Alpha Vantage to enable

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automatic retrieval and structured parsing of market quotes, news, fundamentals, and market sentiment.

- Designed state management and memory mechanisms: implemented lightweight historical scenario retrieval based on BM25, combined with a reflection mechanism to form an iterable investment research memory loop.
- Completed end-to-end engineering implementation, supporting automatic execution with specified tickers and dates, and outputting comprehensive decision logs and standardized trading signals

Medical Pathology Report Understanding with LoRA Fine-tuning

04/2026

Overall: A prototype of the pathological report understanding system was constructed based on LLaMA-Factory, Owen3.8B and LoRA, completing the full-process development from data processing, domain continuous pre-training, supervised fine-tuning to local reasoning services and the front end.

GitHub: https://github.com/terzotian/CDS547LLM_LoRA

- ✧ The training data was constructed using TCGA-Reports pathological texts and cancer type labels to complete the domain adaptation training of pathological reports and the fine-tuning of cancer type classification tasks.
- ✧ Implement the locally runnable training and inference processes in the Apple Silicon (MPS) environment and conduct experimental records and visual monitoring through Wandb.
- ✧ Develop an interactive platform that supports single case/multi-case report input, random sample loading, cancer type prediction, and structured pathological summary display.

A Stock Price Prediction and Visualization Analysis System Based on Random Forest and LSTM

03/2026

Background: Based on Python, Scikit-learn and TensorFlow, the OHLCV time series data of stocks were cleaned, feature engineered and predictive modeled. Three types of models, namely Baseline, Random Forest and LSTM, were constructed and cross-validation and error analysis of time series were carried out. Generate over 30 visual charts and indicator summaries

GitHub: <https://github.com/terzotian/XGboost>

- ✧ Responsible for the end-to-end modeling process design and implementation of the stock price prediction project, completing data cleaning, feature engineering, random forest and LSTM model training, time series cross-validation and error analysis based on Pandas/Scikit-learn/TensorFlow
- ✧ Derivative features such as days of the week, months, and spreads are constructed based on OHLCV financial time series data. The robustness of the model is evaluated using time sequence segmentation and 50% TimeSeriesSplit, and visualization results such as feature importance, error distribution, and prediction fitting are generated
- ✧ The model selection was completed through the comparative experiments of Baseline, Random Forest, and LSTM. Among them, Random Forest achieved RMSE 1.394, MAE 0.975, and R^2 0.9967 on the test set, forming indicator reports and reusable analysis scripts

A Sentiment analysis pipeline for English comments on YouTube based on PySpark

04/2026

An end-to-end three-class sentiment analysis pipeline for YouTube English comments (data cleaning → feature engineering → multi-model training/parameter tuning → fusion → visualization and model disk writing) was built using PySpark, and comparative experiments and improvements were conducted on multiple types of representations such as TF-IDF, Word2Vec, GloVE, and BERT.

GitHub: https://github.com/terzotian/CDS527_PySpark_Sentiment_Analysis

- ✧ Build a three-category sentiment analysis Pipeline for YouTube comments based on PySpark, covering robust parsing of multi-line CSV, cleaning and deduplication, word segmentation/stop word removal, feature engineering, model training and evaluation, and disk reuse (Spark ML Pipeline).
- ✧ Implement and compare multiple text representations such as TF-IDF +LR, Word2Vec, GloVE (broadcast+UDF sentence vectors), BERT (extracting sentence vectors from Transformers and reinjecting them into Spark), and DT/ONEVsREST-GBT classifiers.
- ✧ Conduct hyperparameter search (including fast mode control resources) through CrossValidator+ParamGridBuilder, and output indicators such as Accuracy/ weightheighted - F1 along with visual comparison charts.
- ✧ The design feature fusion (TF-IDF +Embedding concatenation) was integrated with the hard voting of the three models, which was improved from the baseline weighted-F1 0.5727 to 0.6837 (hard voting), and the improvement report and reproducible experimental log were accumulated.

Kronos Multi-Agent Forecasting & Trading (Kronos-based)

01/2026

Overall: A generative prediction system for financial Time series (OHLCVA) is built around the core of "Tokenizer quantization and discretization + Transformer autoregression generation + multi-pipeline fine-tuning + Web visualization", and it has a complete basic capability of "training - inference - visualization - backtesting".

Core technical architecture:

- ✧ The core of sequence modeling: By combining "binary spherical quantization (BSQuantizer) + hierarchical Transformer", continuous OHLCVA sequences are discretized into tokens and then autoregressive generation is performed, considering both compression efficiency and generation capability. The Predictor layer introduces

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dedicated time series designs such as RoPE self-attention, cross-attention conditioning, and time embedding to adapt to the time dependence of financial sequences.

Flexible training system:

- ✧ Supports dual pipelines (Qlib professional quantization/CSV universal), multiple training modes (sequential/single component/distributed), and lightweight fine-tuning (LoRA for attention/FFN layer), which can adapt to different data sources, computing power environments and adaptation requirements.

Engineering foundation: Covering multiple devices such as CPU, CUDA, and MPS, it provides Web UI visualization and regression testing to ensure stable output and has an end-to-end basic link from "model training" to "policy backtesting".

The core logic of optimization:

- ✧ Shifting from a "model-centric" approach to a "trade-centric" one—no longer solely pursuing point prediction accuracy, but ensuring prediction results can be transformed into stable, tradable, and verifiable profits. Specifically, outline the phased implementation value and implementation paths.

Driver Fatigue Monitoring System Based on Computer Vision

11/2025

The overall system architecture adopts a three-layer modular structure of "Data - Model - Decision", forming a complete closed loop of "Data Collection - Feature Extraction - Fatigue Prediction - Intelligent Decision-Making - Feedback Optimization", ensuring seamless connection and efficient collaboration among all links.

- GitHub: <https://github.com/S1M0nLEE/Foundation-of-AI>
- The design concept of the architecture: The three-layer architecture performs its own duties while also interacting with each other:
 - ✧ The data layer is responsible for precise collection and feature extraction, providing high-quality input for subsequent analysis.
 - ✧ The model layer is based on the fusion of multiple algorithms to achieve precise prediction of fatigue levels.
 - ✧ The decision layer, making level dynamically optimizes intervention strategies through reinforcement learning to balance safety assurance and driving experience.
- Core workflow data starts from collection. After multi-dimensional features are extracted by **MediaPipe**, they are input into the random forest or CNN model for fatigue level determination. Then, the **Q-Learning** algorithm generates personalized intervention instructions based on the determination results. Finally, the intervention is executed through hardware, and the feedback data is fed back to the system. Achieve continuous optimization of the model.

A Personal Resume Website

04/2026

- Determined the placement and styling of the website
- Used React Framework and HTML, CSS, JavaScript and PHP codes to generate dynamic displays and adaptive layouts
- Uploaded website codes onto the Firebase server to realize online connection and gain a domain name and an email account, enabling user access and contact
- Link: <https://terzoresume.web.app/>

Wildfire Smoke Detection Software Based on YOLOv8 Part of "An Intelligent UVA Used for Early Forest Fire Detection"

06/2023

Member (responsible for Fire-Smoke generation)

Introduction: The system enables an all-round detection of forest fire by connecting the Himawari-8 data server through the FTP connection pool, a fire lookout tower, and a drone with camera.

- Engaged in team discussions to determine the specific goals of the algorithm, including accuracy requirements and application scenarios
- Referred to similar cases, compared varied algorithms for wildfire smoke detection, selected the YOLOv8 deep learning model, and examined its pros and cons
- Applied the YOLOv8 model to work out the algorithm, including:
 - ✧ Collected datasets containing images of wildfire smoke, preprocessed the collected data and identified relevant features for smoke detection
 - ✧ Defined model parameters, trained the model with the annotated data, and optimized its performance by adjusting hyperparameters
 - ✧ Assessed the accuracy and generation capability of the model and modified it based on validation results
 - ✧ Evaluated the final model to measure its performance in actual scenarios
- Achievement: The application for software copyright was submitted to the China Copyright Protection Center.

A Front-End and Back-End Separated System Based on Seafood Price Prediction

04/2025

- My undergraduate thesis was a front-end and back-end separated project developed based on the Vue and Django frameworks. Using Jupyter notebook, I crawled and cleaned the aquatic product data from Wanbang International Data website and saved it in the Navicat database.
- The web system's function is to provide registered users with a list of aquatic products sorted by time. The administrator is responsible for managing users and products. Users can interactively view the price changes of different aquatic products in each province, the price comparisons among products, and the visualization of the origin

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and the quantity of each province; with the help of machine learning, they can predict price trends.

A Decision Tree Model for Prediction

05/2024

- Accessed the Titanic survival dataset online and preprocessed the dataset with Jupyter notebook, including deleting irrelevant data and converting the non-numeric data to numeric data
- Created new features using feature engineering techniques and presented data with visual charts or tables
- Selected suitable algorithms to train the model and adjusted hyperparameters to improve model performance

COMPETITIONS

University Computer Games Championship

➤ Gomoku Competition, Leader

05/2024-08/2024

Supervisor: Dr. Jiang Changjin

Introduction: The University Computer Games Championship includes traditional games, such as chess, Go, and Gomoku. Gomoku competitors are required to develop an AI program for playing Gomoku to match against others.

- Led discussions to determine the required functions of the program and search algorithms needed for program development, and planned the design of the evaluation function and user interface
- Mastered Gomoku rules and conceived how to design opening functions in accordance with Yamaguchi opening rule
- Applied Alpha-Beta pruning and heuristic search to increase the efficiency of identifying optimal moves
- Tested and improved the C++ program in Qt Creator, perfected the user interface, and optimized opening functions to provide professional game experiences
- Achievement: won the national first prize with the AI Gomoku software designed with a 15×15 Go board

➤ No Go Competition, Member

05/2024-08/2024

Introduction: No Go Competition in the Championship, completely different from normal Gomoku games, allows competitors to place stones in turns. The aim is not to eat the other player's stones, or otherwise the player placing stones lose a score. Suicide and pass are prohibited, and there is no draw in this competition.

Supervisor: Dr. Jiang Changjin

- Tested and modified possible program failures
- Prepared the technical report on the gaming software, especially the UB-tree, UCB algorithm, and move generator used in the software program
- Achievement: won the national first prize with the AI gaming software for No Go designed with a 9×9 Go board

INTERSHIPS

Synnex Technology International Corporation (Shenzhen)

09/2024-01/2025

Huawei Datacom Pre-sales Engineer

- Support the pre-sales technical work of Huawei's data communication products. Based on the customer's network architecture, business traffic and security requirements, complete the model selection of products such as switches, routers and firewalls, and design customized network deployment solutions. Independently write over 3 standardized technical solution documents.
- We have independently output over 10 sets of customized network solutions, efficiently supporting the implementation of customer demands and the early advancement of projects. Through high-frequency customer connection and demand decomposition, quickly master the core features and networking architecture of data communication products, achieve a full-process capability breakthrough from demand understanding → solution design → customer communication and Q&A → solution optimization. The efficiency of technical presentation and cross-party communication has been increased by 20% and possess the ability to output technical solutions and gain insights into customer demands

Bank of Communications, Zhuhai

07/2024-08/2024

Customer Service Manager Assistant

- Assist bank relationship managers in conducting customer relationship maintenance and business support work, familiar with the full process of banking business and customer management specifications.
- Responsible for organizing customer information, filing materials, and entering data into business systems to ensure 100% accuracy and efficiency of information; proactively connected with customer needs, responded promptly, and coordinated to handle and resolve urgent customer issues, supporting the team in promoting business implementation.
- Demonstrated strong initiative, cross-departmental collaboration skills, and service awareness during work, effectively improving customer management efficiency by 30% and issue response speed.

Zhuhai New Horizon Communication Development Co., Ltd.

07/2022-08/2022

Intern

- Completing the full process management of product delivery, planning delivery plans according to order requirements, coordinating logistics arrangements and progress tracking, ensuring efficient and timely delivery for 7+ business.
- Responsible for the systematic organization, filing, and maintenance of customer information systems, establishing standardized information ledgers, and improving customer data management efficiency by 40%; cooperate with

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colleagues to complete order docking, demand response, and after-sales follow-up, strengthening project execution, cross-party coordination, and refined operational capabilities.

QUALIFICATION

Huawei Developers

08/2024

- Successfully completed the online course on Harmony OS application development and received a pass

Cisco Systems, Inc.

11/2022

- Attended online lectures pertinent to cybersecurity, including cybersecurity administration, threat analysis, system safeguards, network defence, resource specialist

ACTIVITIES

Class Monitor

09/2021-06/2025

- Assisted with managing student affairs
- Student Member of Chinese Association for Artificial Intelligence
- Achievement: Awarded the title of “Outstanding Class Monitor”